

2.1

```
SELECT
  lib_zone,
  code_zone,
  AVG(code_qual) AS moyenne
FROM qualite_air
WHERE type_zone = 'commune'
GROUP BY lib_zone, code_zone
ORDER BY moyenne DESC
LIMIT 1;
```

=====

2.2

```
SELECT
  lib_zone,
  code_zone,
  code_qual,
  AVG(code_qual) OVER (PARTITION BY lib_zone, code_zone) AS moyenne,
  MAX(code_qual) OVER (PARTITION BY lib_zone, code_zone) AS maximum
FROM qualite_air
WHERE type_zone = 'commune';
```

=====

2.3

```
SELECT
  d.num_dep,
  d.nom_departement,
  AVG(q.code_qual) AS moyenne,
  MAX(q.code_qual) AS maximum,
  RANK() OVER (ORDER BY AVG(q.code_qual) ASC) AS classement
FROM qualite_air q
JOIN commune c ON q.code_zone = c.code_commune
JOIN departement d ON c.num_dep = d.num_dep
WHERE q.type_zone = 'commune'
GROUP BY d.num_dep, d.nom_departement
ORDER BY classement;
```

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2.4

```
SELECT
  c.num_dep,
  d.nom_departement,
  c.nom_commune AS libelle_zone,
  MAX(q.conc_pm10) AS max_par_ville
FROM qualite_air q
JOIN commune c ON q.code_zone = c.code_commune
JOIN departement d ON c.num_dep = d.num_dep
WHERE d.nom_departement = 'Ain' AND q.type_zone = 'commune'
GROUP BY c.num_dep, d.nom_departement, c.nom_commune
ORDER BY max_par_ville DESC
LIMIT 1;
```

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2.5

```
WITH ClassementVilles AS (  
  SELECT  
    c.num_dep,  
    d.nom_departement AS nom_dep,  
    c.nom_commune AS libelle_zone,  
    MAX(q.conc_pm10) AS max_pm10_dep,  
    RANK() OVER(PARTITION BY c.num_dep ORDER BY MAX(q.conc_pm10) DESC) as  
rang  
  FROM qualite_air q  
  JOIN commune c ON q.code_zone = c.code_commune  
  JOIN departement d ON c.num_dep = d.num_dep  
  WHERE q.type_zone = 'commune'  
  GROUP BY c.num_dep, d.nom_departement, c.nom_commune  
)  
SELECT  
  num_dep,  
  nom_dep,  
  libelle_zone,  
  max_pm10_dep  
FROM ClassementVilles  
WHERE rang = 1  
ORDER BY num_dep;
```

=====

2.6

```
SELECT  
  c.num_dep,  
  d.nom_departement,  
  SUM(CASE WHEN q.code_qual = 1 THEN 1 ELSE 0 END) AS "nb bon indice",  
  SUM(CASE WHEN q.code_pm10 = 1 THEN 1 ELSE 0 END) AS "nb bon indice de pm10",  
  SUM(CASE WHEN q.code_o3 = 1 THEN 1 ELSE 0 END) AS "nb bon indice de o3"  
FROM qualite_air q  
JOIN commune c ON q.code_zone = c.code_commune  
JOIN departement d ON c.num_dep = d.num_dep  
WHERE q.type_zone = 'commune'  
GROUP BY c.num_dep, d.nom_departement  
ORDER BY c.num_dep;
```

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2.7.

WITH ComptageIndices AS (

SELECT

c.num_dep,

d.nom_departement,

SUM(CASE WHEN q.code_qual = 1 THEN 1 ELSE 0 END) AS nb_bon_indice,

SUM(CASE WHEN q.code_pm10 = 1 THEN 1 ELSE 0 END) AS nb_bon_indice_pm10,

SUM(CASE WHEN q.code_o3 = 1 THEN 1 ELSE 0 END) AS nb_bon_indice_o3

FROM qualite_air q

JOIN commune c ON q.code_zone = c.code_commune

JOIN departement d ON c.num_dep = d.num_dep

WHERE q.type_zone = 'commune'

GROUP BY c.num_dep, d.nom_departement

)

SELECT

num_dep,

nom_departement,

nb_bon_indice AS "nb bon indice",

RANK() OVER (ORDER BY nb_bon_indice DESC) AS rang_bon_indice,

nb_bon_indice_pm10 AS "nb bon indice de pm10",

RANK() OVER (ORDER BY nb_bon_indice_pm10 DESC) AS rang_bon_indice_pm10,

nb_bon_indice_o3 AS "nb bon indice de o3",

RANK() OVER (ORDER BY nb_bon_indice_o3 DESC) AS rang_bon_indice_o3

FROM ComptageIndices

ORDER BY num_dep;